OPERATIONS LEADERSHIP | LEAN SIX SIGMA | KAIZEN | MANUFACTURING ANALYTICS

Standardizing Custom HVAC Fabrication to Create Capacity and Lead-Time Visibility

Laminaire S.A.S. | Production Manager and Process Improvement Lead

ROLE SCOPE
Oversight of two production floors and their operating leaders

PROJECT SCOPE
Custom ductwork operation located on one production floor

DURATION
3-4 week investigation; phased rollout and stabilization over several months

METHODS & SYSTEMS
Lean, Six Sigma, Kaizen, work measurement, Excel, Ofimática

Case snapshot

Sales could not consistently provide defensible delivery dates for custom ductwork because production variability was not translated into engineered timing standards or visible capacity information. Routine estimates depended heavily on the line leader's experience.

I led the investigation, measurement strategy, classification design, pilot sequence, cross-functional coordination, standard-work development, training, and business requirements for the Ofimática system changes.

> **The solution converted custom-product variability into repeatable planning categories, standard times, and exception-based management.**

Tenure and leadership context

I worked one level below the Director of Operations for approximately 18 months. This initiative ran alongside other Lean, Kaizen, logistics, painting, production, and workforce-capacity improvement projects.

Two production floors | Cross-functional leadership | Industrial engineering | Quality systems

Sales gained a live view of production capacity and generally contacted Production only when concessions, priority changes, or unusual constraints required management judgment.

Leadership Context and Project Governance

My operating organization

As Production Manager, I oversaw daily execution and continuous improvement across two production floors:

- Raw-material receiving, inspection, and disposition
- CNC and punch-press cutting, plus line-specific cutting
- Assembly lines organized by product family
- Multiple painting booths and spray operations
- Production-side packing and coordination with central Logistics
- Two floor industrial engineers and one demand planning engineer
- Functional leaders and multiple assembly-line leaders

I also determined when production services or labor should be subcontracted.

Ductwork project team

My role: operational owner and project lead. I performed most of the investigation, analysis, process design, meeting coordination, pilot planning, and implementation support.

Technology partner: the IT Director. The programming team implemented the approved changes in Ofimática; I defined and coordinated the operational requirements.

Operational contributors: demand planning, floor industrial engineers, production supervisors, line leader, operators, Sales, Logistics, and the Director of Operations when validation or decisions were required.

> **Leadership distinction:** I led and designed the operational solution; the IT team built the corresponding system changes.

Retrospective Project-Management Alignment

Initiating
Defined the delivery-estimation problem, project scope, sponsor, stakeholders, and operational stakes.

Planning
Designed the Gemba investigation, measurement plan, stakeholder meetings, classification logic, and pilot sequence.

Executing
Conducted studies, created standards, developed SOPs and videos, trained teams, and coordinated Ofimática changes.

Monitoring & Controlling
Compared calculated baselines with actual performance, reviewed pilots, and adjusted methods and standards.

Closing
Transferred ownership into routine planning, documented the process, and established ongoing performance feedback.

1. Business Challenge

Custom ductwork varied by configuration, dimensional range, material and construction requirements, fabrication route, and complexity. Production did not have a consistent way to translate those differences into category-level standard times and capacity requirements.

Sales relied on the line leader to interpret each order, assess current workload, and provide an experience-based date. The dependency slowed customer responses, limited visibility, and made routine delivery commitments difficult to support consistently.

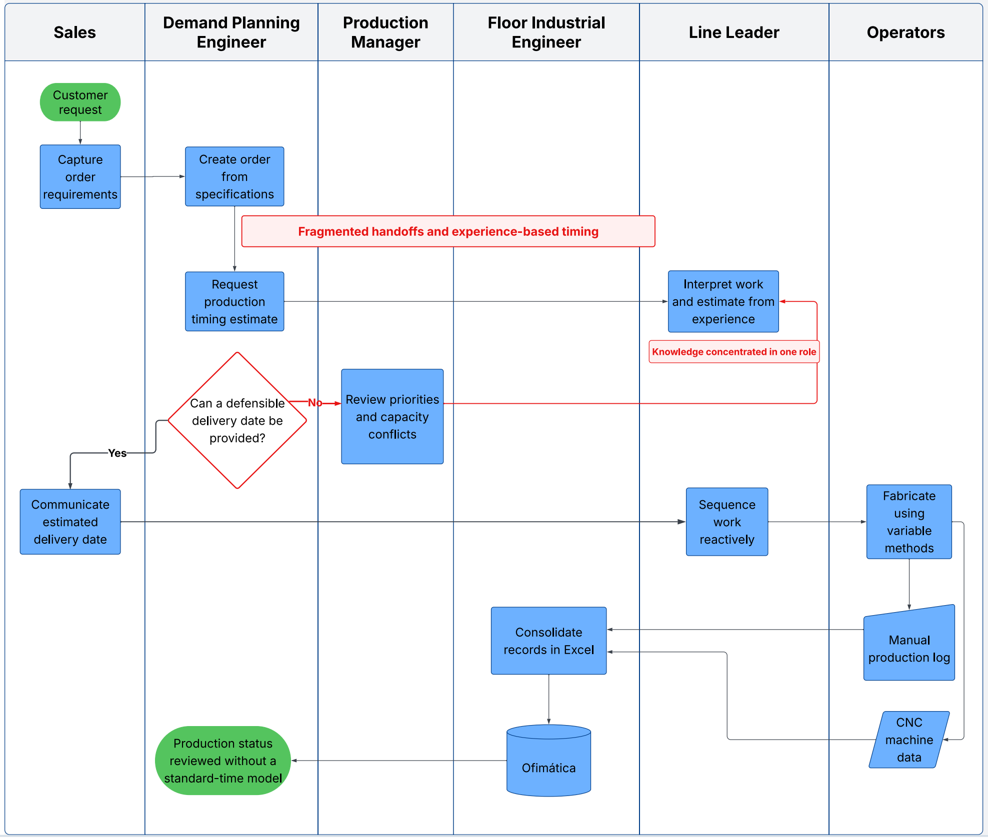
> **Operational stake:** the problem was not customization itself. The problem was unmanaged variability, incomplete standard work, and planning knowledge concentrated in one role.

2. Gemba Investigation

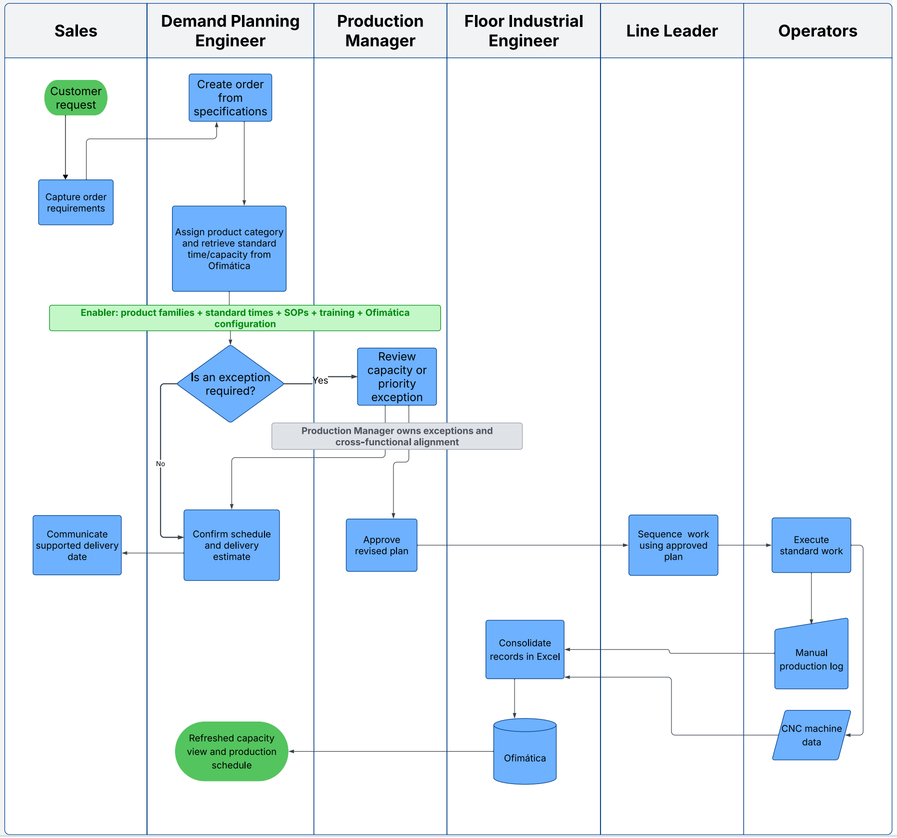
Using Genchi Genbutsu - "Go and See" - I began by studying the actual work. During the first week, I spent approximately five hours per day on the shop floor; the concentrated investigation continued for roughly three to four weeks.

- Followed representative orders through fabrication
- Interviewed operators about methods, variation, delays, and decision points
- Mapped material movement, handoffs, and information flow
- Observed where work depended on the line leader
- Reviewed manual production logs and CNC-generated data

Current-State Cross-Functional Process Map



Future-State Cross-Functional Process Map



3. Measurement and Root-Cause Analysis

Standard Work Measurement Analysis

Translated portfolio excerpt from retained analysis

Product configuration L-C, 14-inch diameter

Production area Accessories

Operators required 1

Total activities 17

Measurement basis Batch and unit time

Activity categories:

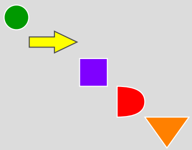
Operation

Transport

Inspection

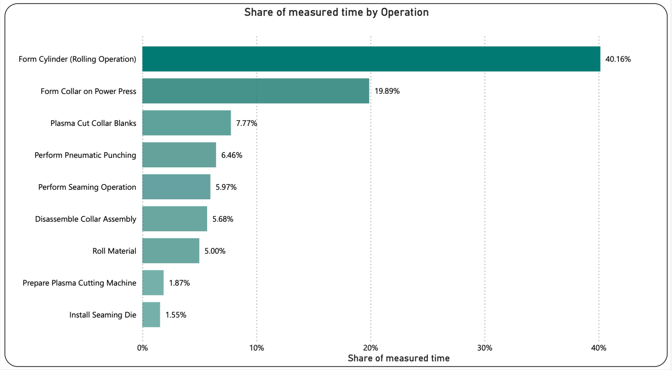
Delay

Storage



Validation Actual line performance + pilot review

Representative Pareto of Measured Work Elements



Method

Repeared stopwatch observations, identification of bottlenecks, delays, and extra movements that did not add value, performance rating, applicable work allowances, statistical sampling, standarization of best practices, and pilot validation.

Representative translated exhibit derived from a retained work-measurement file. Selected labels are generalized for portfolio readability.

Time-and-motion methodology

Representative configurations were decomposed into discrete operations and supporting activities. The production supervisors and I collected repeated stopwatch observations across relevant product and diameter ranges, including material-handling and transportation time as separate categories.

Observed times were evaluated using statistical sampling, performance ratings, and applicable fatigue and personal allowances. Calculated baselines were compared with actual line performance, reviewed against takt and capacity requirements, and tested through controlled pilots.

Validation

- Actual-versus-calculated performance comparison
- Operator and supervisor review
- Committee review with the line leader and Director of Operations
- Pilot runs before broader category rollout

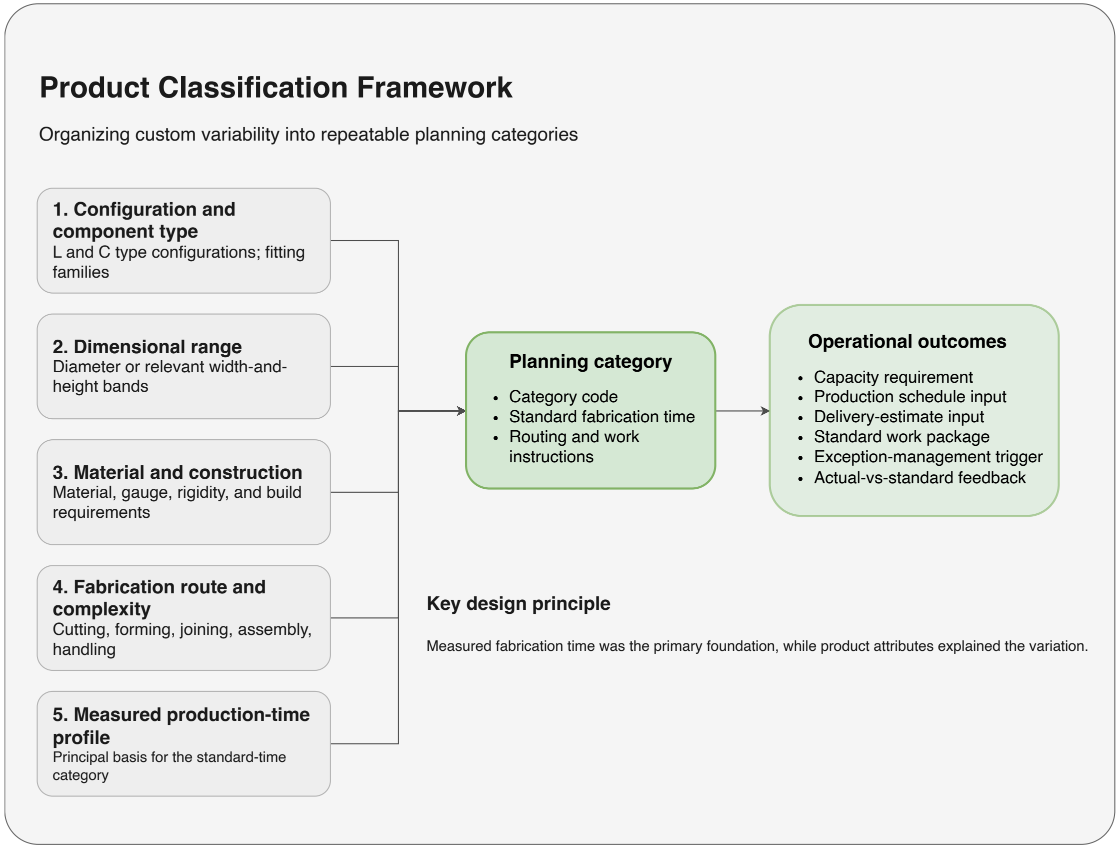
Excel analytical layer: I used macros, pivot tables, advanced calculations, and ranked time analysis to identify high-impact operations and establish repeatable baselines.

INVESTIGATION OUTPUT

Root-cause findings

- No engineered standard times by product configuration
- No consistent classification of product variability
- Incomplete standard work and role definition
- Planning knowledge concentrated in the line leader
- Production status and timing standards were not connected
- Sales lacked direct capacity visibility

4. Solution and Implementation



I converted product variability into a structured planning model. The historical system contained more than four categories and used a combination of:

- **Configuration and component type:** L- and C-type configurations and fitting families
- **Dimensional range:** diameter or relevant width-and-height bands
- **Material and construction requirements:** material type, gauge, rigidity, and build requirements
- **Fabrication route and complexity:** cutting, forming, joining, assembly, and handling steps
- **Measured production-time profile:** the principal basis for assigning the standard-time category

Phased implementation

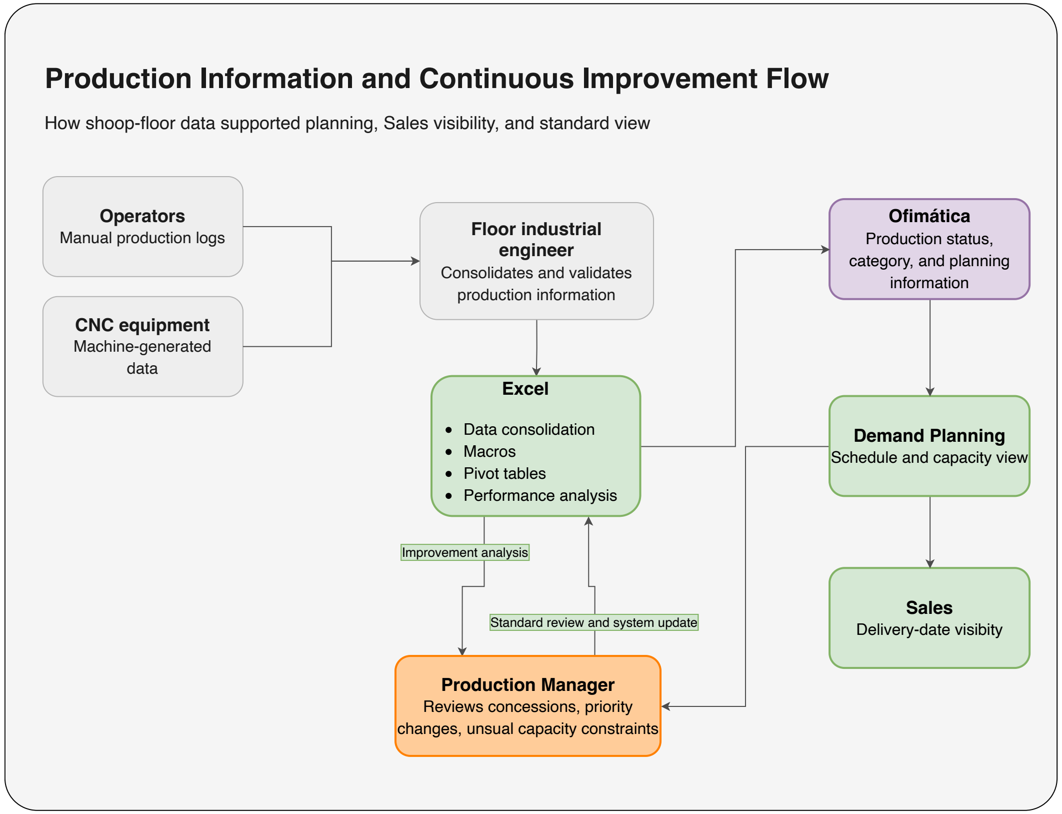
1. Prioritized categories by operational return and feasibility.
2. Piloted one category and compared actual performance with the proposed standard.
3. Reviewed results with the validation committee and adjusted methods, timing, and documentation.
4. Rolled out subsequent categories sequentially.
5. Delivered SOPs, work instructions, training, and how-to videos.
6. Coordinated with the IT Director so programmers could add approved categories and planning logic to Ofimática.

SUSTAINING THE CHANGE

Adoption and change management

Standardization redistributed decision authority and initially raised concerns about workload and job security. I used phased pilots, employee participation in validation, extensive training, video-supported instructions, direct communication, and broader improvements to incentives and benefits to support adoption. The categories entered operational use immediately, followed by a learning curve. Within the next few months, the process operated with substantially less friction and less dependence on informal explanations.

5. Operational Impact



Sales moved from "call the line leader" to a live capacity view supported by standardized planning information.

Worked example

Before: A custom L- or C-type fitting entered planning. The planner needed the line leader to interpret the product, assess workload, and provide an experience-based date.

After: The planner identified the applicable category using configuration, dimensions, construction requirements, and fabrication route. The associated standard time and capacity requirement supported a delivery window visible to Sales.

Exception: I became involved when a concession was requested, priorities shifted, the product fell outside established categories, or capacity required subcontracting.

Capabilities demonstrated

Operations Leadership · Project Management · Lean Six Sigma · Kaizen · Genchi Genbutsu · Process Mapping · Work Measurement · Statistical Sampling · Root-Cause Analysis · Standard Work · Excel Analytics · Business-System Requirements · Pilot Validation · Change Management · ISO 9001 / Quality Systems

DAY-TO-DAY EFFECT

Operational outcomes

- Sales gained live visibility into production capacity and planned delivery information.
- Routine delivery estimates no longer required repeated consultation with the line leader.
- Demand Planning had a repeatable method for translating product characteristics into capacity requirements.
- Production Management became involved primarily for concessions, priority changes, unusual complexity, or subcontracting decisions.
- Knowledge became embedded in categories, standard times, SOPs, work instructions, training videos, Excel analysis, and Ofimática.